20 CONCEPTS · 20 PRACTICE QUESTIONS

Biostatistics Study Guide

Concept review with worked answers, from descriptive statistics to Bowley's skewness.

01

Descriptive Statistics

Summarize observed data with means, medians, percentages, tables and graphs; they describe only the group studied.

A pharmacist averages the ages of the 60 patients seen in one week, only to describe them. What type of statistics is this?

A. Inferential statistics
B. Descriptive statistics
C. Probability sampling
D. Experimental research

Answer: B. The average summarizes the patients observed.

02

Inferential Statistics

Use sample data to estimate population characteristics or test claims, accounting for sampling variability; a biased sample misleads.

A researcher studies 250 patients to estimate medication adherence among all patients at a hospital. Which type of statistics is used?

A. Descriptive statistics
B. Data organization
C. Inferential statistics
D. Frequency counting

Answer: C. The sample estimates adherence in the larger population.

03

Population and Sample

A population is everyone in the research question; a sample is the selected members. A population value is a parameter, a sample value a statistic.

A university has 8,400 students; a researcher selects 140 to study sleeping habits. Which statement is correct?

A. Population is 140, sample is 8,400
B. Both are 140
C. Population is 8,540
D. Population is 8,400, sample is 140

Answer: D. All 8,400 form the population; the 140 form the sample.

04

Steps in Handling Data

Collect with consistent procedures, organize and check, summarize, analyze, then interpret against the question and the study's limits.

Which sequence best describes the handling of statistical data?

A. Collect → Organize → Summarize → Analyze → Conclude
B. Analyze → Collect → Organize → Summarize → Conclude
C. Summarize → Organize → Collect → Conclude → Analyze
D. Collect → Conclude → Analyze → Organize → Summarize

Answer: A. Collected, organized, summarized, analyzed, then interpreted.

05

Data Sets, Variables, Observations

A data set organizes information on study individuals; a variable is a characteristic; one individual's recorded value is an observation. Rows are individuals, columns are variables.

A spreadsheet has one row per patient and columns for age, blood group and body weight. Which statement is correct?

A. Blood group is an observation; the spreadsheet is one variable
B. The spreadsheet is a data set, age is a variable, one patient's age is an observation
C. Each patient is a variable; body weight is a population

06

Measurement Scales

Nominal names categories; ordinal ranks them; interval has equal steps but no true zero; ratio has a true zero. 80 kg is twice 40 kg, but 40°C is not twice 20°C.

Which option correctly matches each variable with its scale?

A. Blood group: ordinal; pain: nominal; °C: ratio; weight: interval
B. Blood group: ratio; pain: interval; °C: nominal; weight: ordinal
C. Blood group: nominal; pain: ordinal; °C: interval; weight: ratio
D. Blood group: interval; pain: ratio; °C: ordinal; weight: nominal

D. Only numerical columns can belong to a data set

Answer: B. Age is the characteristic; a patient's age is its recorded value.

Answer: C. Nominal, ordinal, interval, ratio, respectively.

07

Grouped Frequency Distributions

Numerical values are combined into mutually exclusive intervals that cover every value, with a count per interval. Easier to read, but exact values are hidden.

A table summarizes ages using intervals 20–<30, 30–<40 and 40–<50. What type of distribution is this?

A. Categorical
B. Ungrouped
C. Nominal
D. Grouped

Answer: D. Numerical ages are combined into class intervals.

08

Class Width

Estimate width as (maximum − minimum) ÷ number of classes, then adjust so classes cover all data without overlapping.

The smallest observation is 12 and the largest is 52, with five classes. What is the estimated class width?

A. 8
B. 10
C. 40
D. 64

Answer: A. 8. (52 − 12) ÷ 5 = 40 ÷ 5 = 8

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Relative and Cumulative Frequency

Relative frequency is class frequency ÷ total. Cumulative frequency is a running total. Relative frequencies sum to 100%; the last cumulative value equals n.

Four classes have frequencies 4, 9, 12 and 5. Find the relative frequency of the third class and the cumulative frequency up to it.

A. 12% and 12
B. 40% and 25
C. 30% and 21
D. 40% and 30

Answer: B. n = 30; 12 ÷ 30 = 40%; 4 + 9 + 12 = 25.

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Histograms

Adjacent bars over class intervals; with equal widths, bar height is frequency, so the tallest bar is the most frequent class.

Frequencies are 3 (10–<20), 8 (20–<30), 14 (30–<40) and 5 (40–<50). Which interval has the tallest bar?

A. 10–<20
B. 20–<30
C. 30–<40
D. 40–<50

Answer: C. Its frequency of 14 is the highest.

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Population Mean

μ = ΣX ÷ N. It is the balance point of the data and is sensitive to unusually large or small values.

The entire population is 6, 8, 10 and 12. What is the population mean?

A. 8
B. 10
C. 12
D. 9

Answer: D. 9. (6 + 8 + 10 + 12) ÷ 4 = 9

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Weighted Mean and GPA

Observations contribute by weight. GPA = Σ(grade points × credits) ÷ Σ credits, so more credits means more influence.

Course A: 4 points, 3 credits. Course B: 3 points, 2 credits. Course C: 2 points, 1 credit. What is the GPA to two decimals?

A. 3.33
B. 3.00
C. 3.50
D. 4.00

Answer: A. 3.33. (12 + 6 + 2) ÷ 6 = 20 ÷ 6 ≈ 3.33

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Quartiles

Three cut points divide ordered data into four portions. Q_1, Q_2, Q_3 are the 25th, 50th and 75th percentiles; Q_2 is the median.

Which statement about quartiles is correct?

A. They divide ordered data into three equal parts
B. They divide ordered data into four approximately equal parts, and Q_2 is the median
C. Q_1 is the median
D. Q_3 is the 25th percentile

Answer: B. Four portions; the second quartile is the median.

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Position of the First Quartile

Q_1 position = $(n + 1) \times 0.25$. It gives a position in the ordered data, not the value; fractions need interpolation.

With 15 ordered observations, what is the position of Q_1?

A. Third
B. Fifth
C. Fourth
D. Eighth

Answer: C. Fourth. $(15 + 1) \times 0.25 = 4$

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Calculating Quartiles

Sort ascending, then use positions $(n + 1)/4$, $(n + 1)/2$ and $3(n + 1)/4$, reading values from the ordered list.

Find the quartiles of 11, 3, 8, 15, 5, 9, 1, 13, 7, 10, 4.

A. $Q_1 = 3$, $Q_2 = 7$, $Q_3 = 10$
B. $Q_1 = 5$, $Q_2 = 9$, $Q_3 = 13$
C. $Q_1 = 4$, $Q_2 = 9$, $Q_3 = 11$
D. $Q_1 = 4$, $Q_2 = 8$, $Q_3 = 11$

Answer: D. Sorted: 1, 3, 4, 5, 7, 8, 9, 10, 11, 13, 15. Positions 3, 6, 9 give 4, 8, 11.

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Interquartile Range

$IQR = Q_3 - Q_1$, the spread of the middle 50%. Less sensitive to extremes than the range or standard deviation.

If $Q_1 = 6$, $Q_2 = 10$ and $Q_3 = 15$, what is the IQR?

A. 5
B. 9
C. 4
D. 21

Answer: B. 9. $15 - 6 = 9$

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Mode

The most frequent value or category; there can be several. It depends on repetition, not size, so extreme values may leave it unchanged.

The data are 2, 2, 5, 5 and 9. If 9 is replaced by 90, which statement is correct?

A. The modes remain 2 and 5
B. The only mode becomes 90
C. There is no mode
D. The modes become 5 and 90

Answer: A. Both 2 and 5 still occur twice, the highest frequency.

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Variance

The average squared deviation from the mean. Population variance divides by N, sample variance by $n - 1$; units are squared, and the square root is the standard deviation.

Population A is 9, 10, 11. Population B is 1, 10, 19. Which statement is correct?

A. A has the larger variance
B. Both are equal because the means are equal
C. B has the larger variance
D. Both have zero variance

Answer: C. Both means are 10, but B deviates far more.

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Skewness

Positive skew means a longer right tail, negative a longer left tail. A normal distribution is symmetric with zero skew; a typical positive skew pulls the mean above the median.

Which statement best describes skewness?

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Bowley's Coefficient of Skewness

$(Q_3 + Q_1 - 2Q_2) \div (Q_3 - Q_1)$. Positive when the upper quartile distance is larger; resistant to extremes; zero does not prove symmetry.

A data set has $Q_1 = 4$, $Q_2 = 7$ and $Q_3 = 12$. What is Bowley's coefficient?

A. A normal distribution has positive skewness
B. A positively skewed distribution has a longer left tail
C. A normal distribution always has mean greater than median
D. A normal distribution has zero skewness; in a typical positively skewed one, the mean exceeds the median

Answer: D. A long right tail typically pulls the mean upward.

A. −0.25
B. 0.25
C. 0.50
D. 0

Answer: B. 0.25. (12 + 4 − 14) ÷ 8 = 2 ÷ 8 = 0.25